

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Pseudocode Writing Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO5 – Naturalization (BL5, BL6)	Assessment:	Assessment Tool 2 – Pseudocode Writing Task
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	Analyze computational problems using backtracking techniques and complexity classes such as P, NP, NP-Hard, and NP-Complete for combinatorial problem solving.		
Student Name:	Sairaj C	Reg. No.:	192411018
Section / Batch:	2024	Date:	25-08-26

Instructions to Students

- Answer the following question with proper pseudocode and logical explanation.
- Clearly specify the algorithmic approach and complexity class wherever applicable.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) for complexity analysis.
- Include state-space representation, recursion steps, and pruning conditions in backtracking problems.
- Neat presentation and step-by-step justification are mandatory.

Question Type	Focus Area	Questions
Pseudocode Writing Task	Analyze combinatorial problems using backtracking and complexity classes such as P, NP, NP-Hard, and NP-Complete.	Q1

SECTION A – Pseudocode Writing Task

ASSESSMENT- CO5-AT2-Pseudocode Writing Task (Algorithm Design)

← Back

Language: Python C C++ Java

QUESTIONS

Q1

Edge Coloring Problem

100 marks

1/1 answered

Q1 Edge Coloring Problem

100 marks

Write a pseudocode using Backtracking to color edges without conflicts. Clearly define inputs (graph) and output (edge colors). Design algorithm ensuring adjacent edges differ. Use control structures effectively. Apply pruning for invalid assignments. Ensure clarity. Handle all cases.

Your Code

```
1 class EdgeColoring:
2     def __init__(self, vertices, edges, k):
3         self.V = vertices
4         self.edges = edges
5         self.k = k
6         self.num_edges = len(edges)
7         self.edge_colors = [-1] * self.num_edges
8     def is_safe(self, edge_idx, color):
9         """
10        Checks if it is safe to assign 'color' to the edge at 'edge_idx'
11        An edge is safe if no adjacent edge (sharing a vertex) has the
12        """
```

Input

stdin input

Output

```
Successfully colored edges using
3 colors:
Edge (0, 1) -> Color 1
Edge (1, 2) -> Color 2
Edge (2, 0) -> Color 3
```

Code of Conduct

I Sairaj C (192411018) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sairaj C

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%				
Algorithm Design	30%				
Use of Control Structures	20%				
Pseudocode Structure	10%				
Completeness	20%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____